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Ex: 7a

DES Algorithm

Aim:

To implement DES algorithm.

Algorithm:

Header Module:

1) Initialize openssl legacy and default providers.

2) Create cipher context using `EVP_CIPHER_new()`

3) Initialize DES ECB mode with `EVP_EncryptInit` using 8 byte shared key.

4) Perform block processing via `EVP_EncryptUpdate` and finalize with padding using `EVP_EncryptFinal-ex`.

Server Module:

1) Create a TCP socket, bind to port 8080, and listen for incoming connections.

2) Accept connection from the client and receive the raw ciphertext bytes over the socket.

3) Decrypt received payload using `DES_decrypt()` and display plain message.

Client Module:

- 1) Create a TCP socket and connect to server address 127.0.0.1 : 8080.
- 2) Encrypt the plaintext using DES-Encrypt with the pre-shared 8-byte key.

Output:

server:

Waiting for incoming client on port 8080

Client connected

Received 48 encrypted bytes

Decrypted message: "Top Secret Payload using des-cipher.h"

Client:

Plaintext: "Top Secret Payload using des.h"

Ciphertext: 8fa2394c8b0e12d502931a20fed87e2
4b010c2cef981249c0d12ccf

Encrypted Ciphertext successfully sent to server.